

# SHADI SEAIDOUN

## SOFTWARE ENGINEER | AI/ML

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### SUMMARY

Computer Science graduate with a strong foundation in software engineering, algorithms, and artificial intelligence. Experienced in applying AI/ML concepts such as neural networks, data preprocessing, and model evaluation through academic coursework and personal projects. Skilled in Python, C/C++, and SQL, with practical experience using AI/ML libraries including Pandas, NumPy, Scikit-learn, and PyTorch. Trilingual communicator (English, French, Arabic) with proven leadership, collaboration, and problem-solving abilities in both academic and professional environments.

### TECHNICAL SKILLS

- Python
- C++
- C
- Java
- SQL
- Pandas
- NumPy
- Scikit-Learn
- Pytorch
- Neural Networks
- Git/Github
- OpenGL
- GLM / GLFW
- MS 365
- Power Automate

### SOFT SKILLS

- Problem Solving
- Adaptability / Flexibility
- Time Management
- Collaboration
- Leadership

### PROFESSIONAL EXPERIENCE

#### Team Lead / Cloud Data Analyst • Shared Services Canada (Remote)

Jul 2022 – Apr 2024

- Led a 5-member student team with weekly check-ins to track progress and resolve challenges
- Automated business requests by extracting data from BITS into Excel and pushing rows into SharePoint lists with automated email alerts.
- Streamlined candidate onboarding by creating Microsoft Forms workflows that generated SharePoint folders, organized documents, and updated MS Lists for real-time tracking.

#### Software Engineer • Sunrise Wholesale Produce (Remote)

May 2024 - Apr 2025

- Developed Python-based delivery route optimization algorithms integrated with Google Maps API for real-time traffic.
- Automated inventory tracking with Python and SQL to monitor stock levels and trigger restocking alerts.
- Enhanced operational efficiency by reducing delivery overhead and preventing supply chain disruptions.

### PROJECTS

#### 3D Path Planning for NP-Hard Shortest Path Problem

- Implemented and compared three fundamental algorithms for the NP-hard problem of 3D shortest path planning
- Developed a parallelized version achieving 76% faster computation while maintaining path quality
- Built an interactive visualization system with C++, OpenGL and ImGui for real-time performance monitoring across varying obstacle densities (1-500)

#### Chest X-Ray Pneumonia Detection using CNN

- Implemented and compared three fundamental algorithms for the NP-hard problem of 3D shortest path planning
- Applied data augmentation, preprocessing, and evaluation metrics (accuracy, precision, recall, ROC/AUC).
- Used Kaggle dataset of 5,863 images (NORMAL vs PNEUMONIA) with training, validation, and test splits.

### EDUCATION

#### The University of Western Ontario

Sept 2021 - Apr 2025

B.Sc. in Computer Science: Honours Specialization in Computer Science, Minor in Mathematics

Relevant Coursework: Machine Learning/AI, Operating Systems, Computer Networks, Parallel and Distributed Computing, Databases, Calculus, Linear Algebra, Probability and Statistics